



Profile

Backend-focused M.Sc. AI student with experience in enterprise application portfolio governance, web crawling, research computing, and data-intensive software systems. Strong interests include backend engineering, RAG/analytics pipelines, and practical machine learning.

Education

- 10/2026–Present **M.Sc. Artificial Intelligence**, *Friedrich-Alexander-Universität Erlangen-Nürnberg*, Erlangen
2023–2026 **Bachelor of Data Science**, *Katholische Universität Eichstätt-Ingolstadt*, Ingolstadt
2019–2023 **IT Specialist**, *Mikołaj Kopernik Telecommunications Technical School*, Poznań

Experience

- 03/2026–Present **Working Student – IT Architecture Support**, SIEMENS, Germany
- Improved the quality and currency of the LeanIX application portfolio by reviewing and correcting incomplete or outdated architecture information.
 - Contacted Mendix application managers to create and update architecture diagrams, keeping solution views and dependencies aligned with the current IT landscape.
 - Evaluated enterprise AI and automation platforms, including Altair, Microsoft Copilot Studio, UiPath, and KNIME, and documented candidate use cases and findings.
- 01/2026–Present **Student Assistant**, KATHOLISCHE UNIVERSITÄT EICHSTÄTT-INGOLSTADT, Germany
- **AI chatbot crawler:** Solely developed and maintained the Python crawling and ingestion pipeline behind KU's student-facing RAG chatbot, replacing a failing scraper and producing AnythingLLM-compatible JSONL from German and English `ku.de` content.
 - **Pipeline and ingestion:** Built a checkpointed crawl, PDF fetch, OCR, and deduplication workflow with sitemap discovery, whitelist-based extraction, SQLite state, resumable incremental recrawls, and hash-aware AnythingLLM REST updates.
 - **QA and delivery:** Audited coverage against the sitemap and CMS page tree, generated error/content-quality reports, added render audits and network-free tests, and containerized the documented pipeline with Docker for handover.
 - **Student research assistant:** Supported the Geography research group with HPC and research workflows on the FAU cluster (Linux/SLURM), including access setup, data transfer, execution support, and troubleshooting.
 - **Documentation:** Wrote user-facing guidance for SSH access, remote execution, and cluster usage for non-technical researchers.
 - **Conference website:** Developed and helped host and maintain the SampTA conference website, supporting updates and ongoing web operations.
- 08/2025–10/2025 **Data Science Intern**, KIMBALL ELECTRONICS, Poland
- Supported IT operations and contributed to database development through SQL optimization, triggers, and stored procedures.
 - Analyzed scrap/repair, forecasting, and packaging datasets; built Repair vs. Scrap and TopLineForecast ML prototypes.
 - Designed a Repair vs. Scrap prediction app proof of concept, improved trigger/workflow reliability, and presented work in Agile teams using Azure DevOps/Git.

Nuremberg – Germany

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Skills

Programming	Python, TypeScript, SQL, JavaScript
Web	React, Next.js, Node.js, FastAPI, Flask, REST APIs
Data & ML	Pandas, NumPy, Scikit-learn, SciPy, Model Evaluation, Matplotlib, Seaborn, Plotly
Pipelines	Web Scraping, RAG Data Pipelines, BeautifulSoup, lxml, Playwright, SQLite, pytest, CLI Tooling, PDF/OCR Processing, AnythingLLM
Enterprise	Application Portfolio Management
Data & Tools	PostgreSQL, SQL Server, Redis, Docker, AWS, Azure DevOps, HPC (SLURM), PowerBI, ETL Pipelines, Data Warehousing
OS	Linux (Bash, SLES), macOS, Windows
Certification	SLE201 SUSE Linux Enterprise Server Administration (SUSE Academic Program, 09/2021)

Projects

Detecting Hallucinations in Large Language Models Using Semantic Entropy [Code]

- Bachelor's thesis comparing semantic entropy with simpler uncertainty scores on three open LLMs.

PageRank-Based Analysis of Scientific Papers [Code]

- Built a scalable PageRank framework for analyzing research influence across millions of papers.
- Developed a distributed Semantic Scholar crawler with PostgreSQL/Redis storage, fault-tolerant task handling, and monitoring.
- Engineered a custom HPC PageRank workflow with monitoring via FastAPI.

MSIT-Hotel-Booking [Live] [Code]

- Built a bilingual hotel booking platform with customer and admin flows for a bleisure-focused property concept.
- Implemented booking lifecycle, waitlist, stay-management, occupancy, and revenue dashboard flows.
- Used Next.js, React, TypeScript, Supabase, and PostgreSQL with row-level security.

Personal Portfolio Website [Live] [Code]

- Designed and built a responsive React/TypeScript/Vite portfolio for experience, projects, skills, and a downloadable CV.
- Structured reusable, content-driven components and sections for a polished developer-facing presentation.

KULTour [Code], Hackathon Winner (Best Open Source)

- Built a travel recommendation app in a 24h hackathon (Tourism Technology Festival, Salzburg).
- Implemented ML-based tourism personalization with an interactive map UI for route and point-of-interest exploration.

Activities & Involvement

Data Science Student Council

- Contributed to student representation and communication around program-related topics.
- Helped organize student events and supported council activities within the program.

Languages

Languages **Polish** (Native) ■ **English** (C1) ■ **German** (A2)

Nuremberg – Germany

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